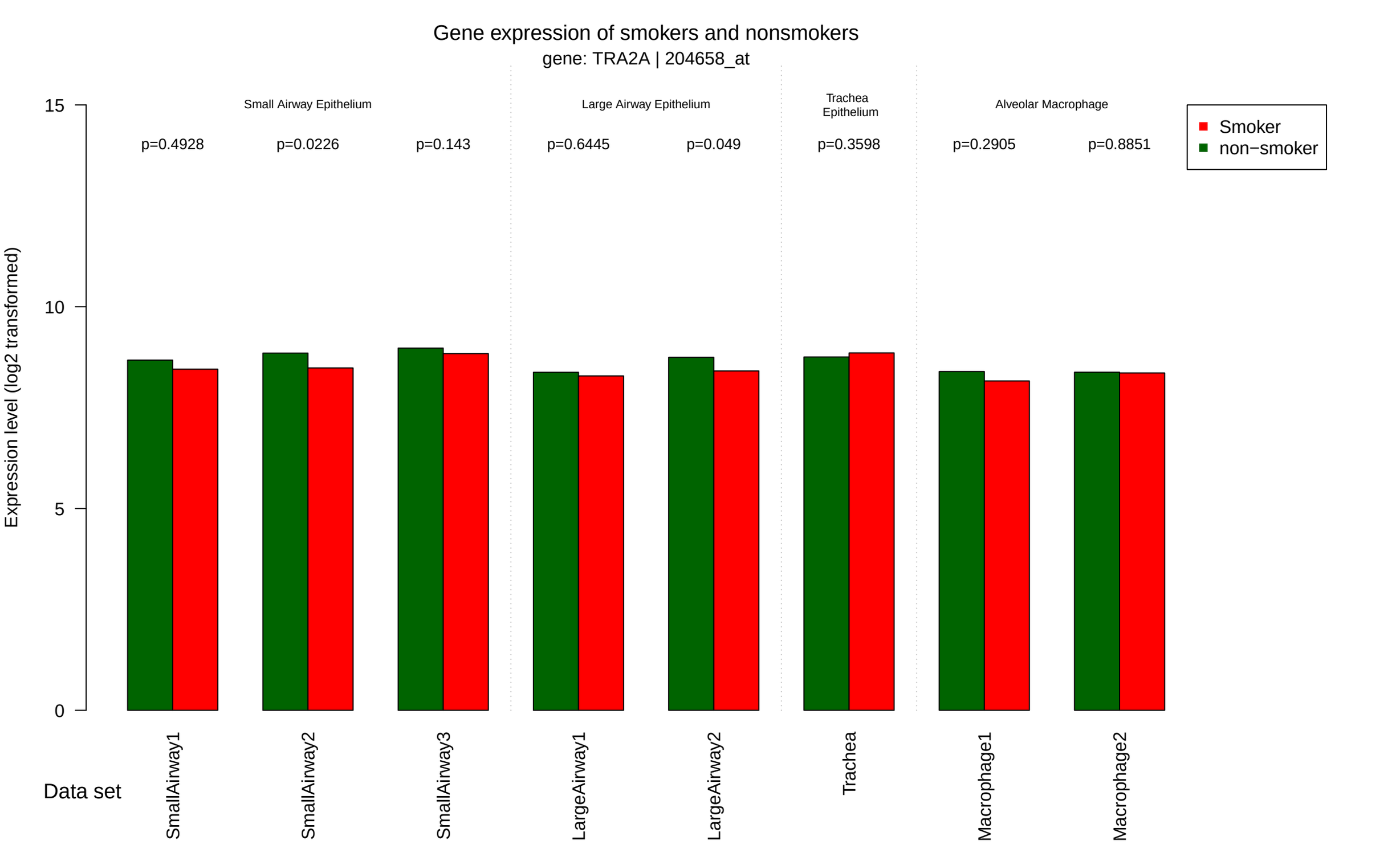
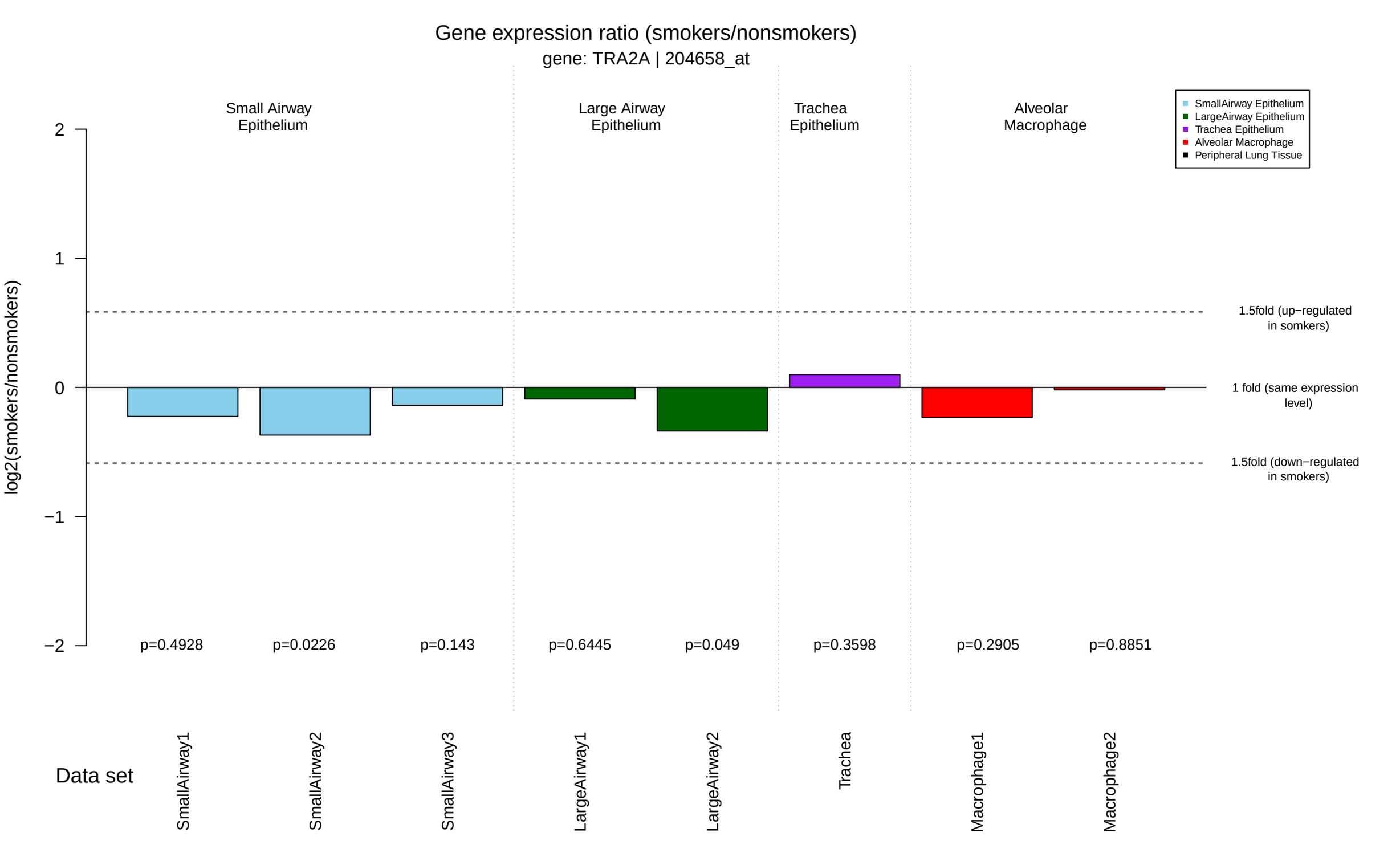
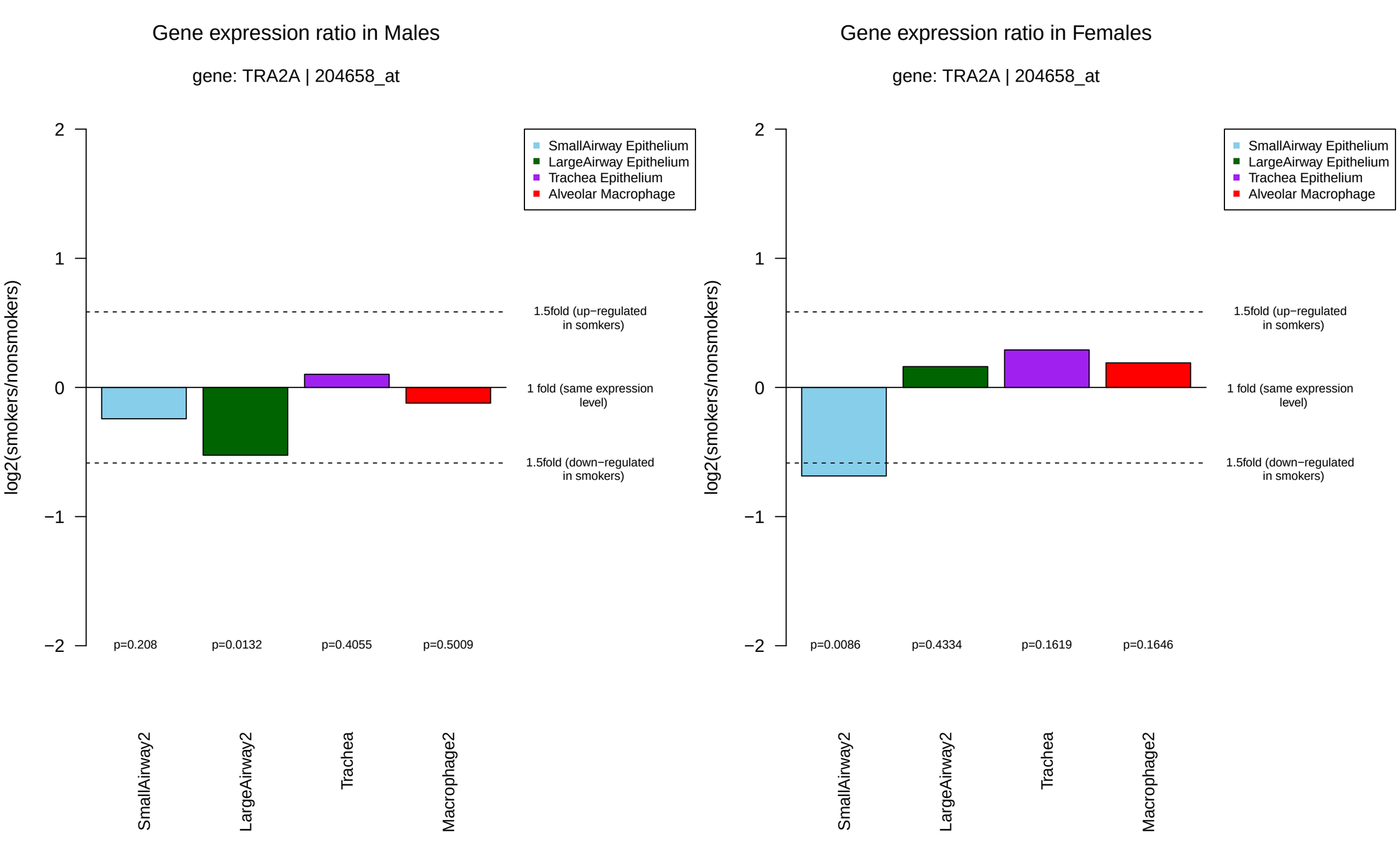
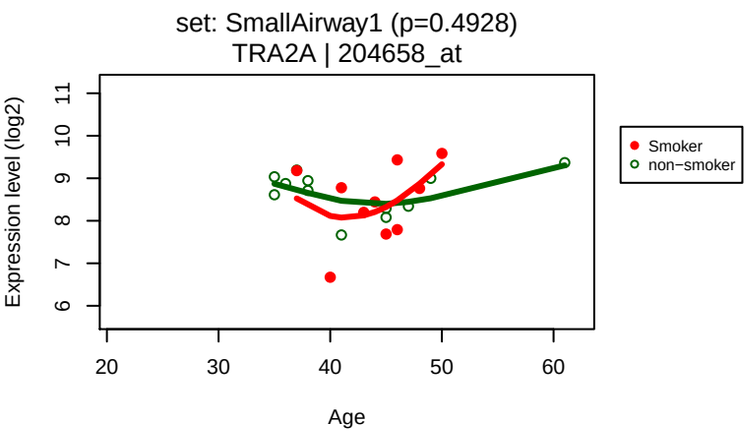
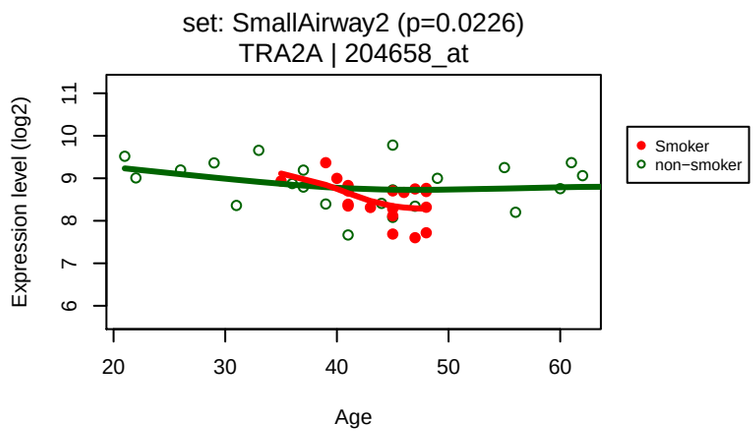
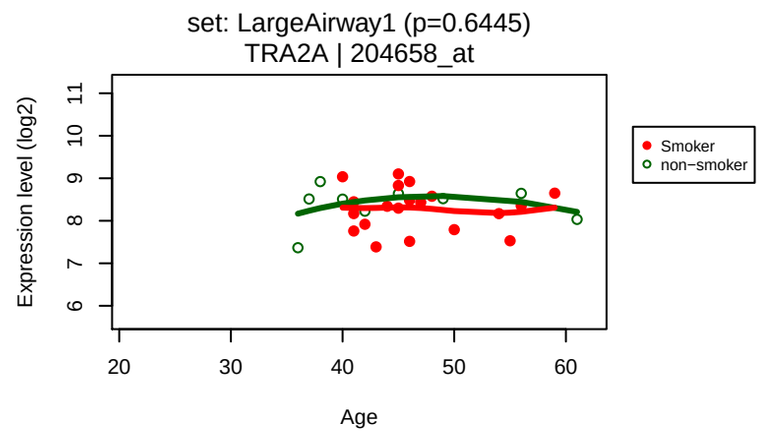

Gene expression of smokers and nonsmokers
gene: TRA2A | 204658_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4928
p=0.0226
p=0.143
p=0.6445
p=0.049
p=0.3598
p=0.2905
p=0.8851
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

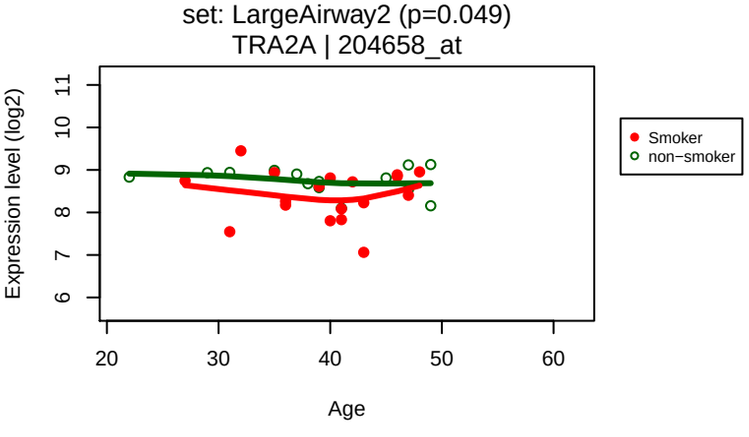

Gene expression ratio (smokers/nonsmokers)
gene: TRA2A | 204658_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4928
p=0.0226
p=0.143
p=0.6445
p=0.049
p=0.3598
p=0.2905
p=0.8851
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

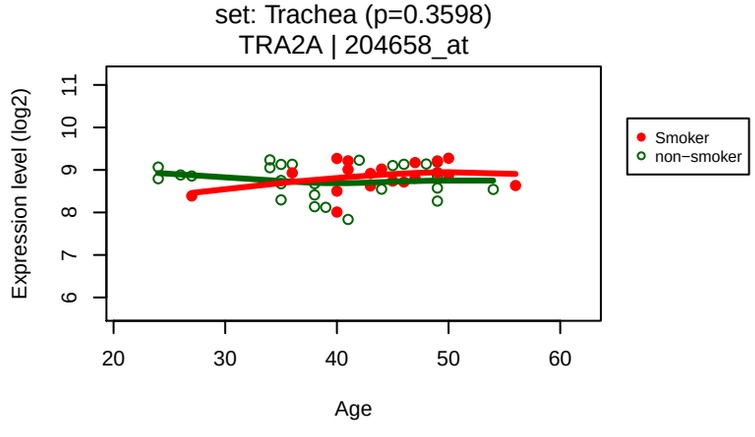

Gene expression ratio in Males
gene: TRA2A | 204658_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.208
p=0.0132
p=0.4055
p=0.5009
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRA2A | 204658_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0086
p=0.4334
p=0.1619
p=0.1646
SmallAirway2
LargeAirway2
Trachea
Macrophage2

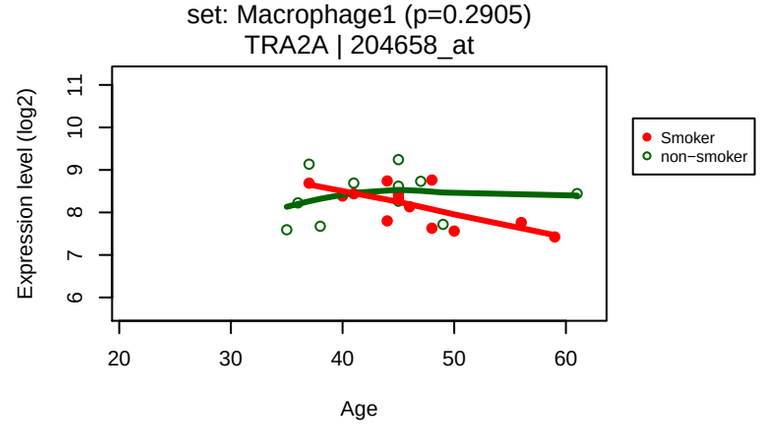

set: SmallAirway1 (p=0.4928)
TRA2A | 204658_at
Expression level (log2)
Age
Smoker
non−smoker

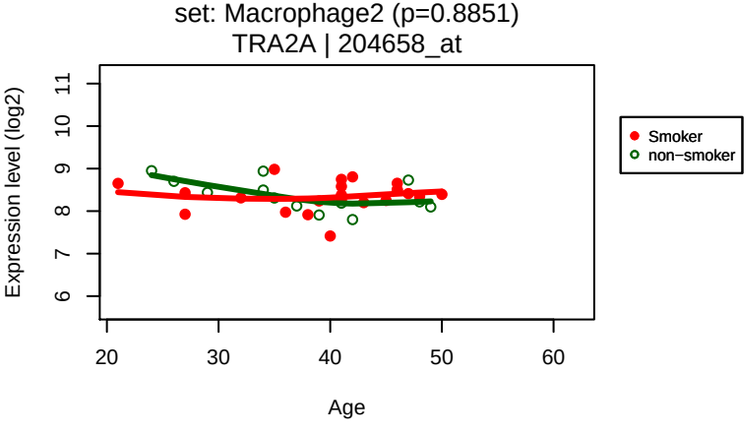

set: SmallAirway2 (p=0.0226)
TRA2A | 204658_at
Expression level (log2)
Age
Smoker
non−smoker

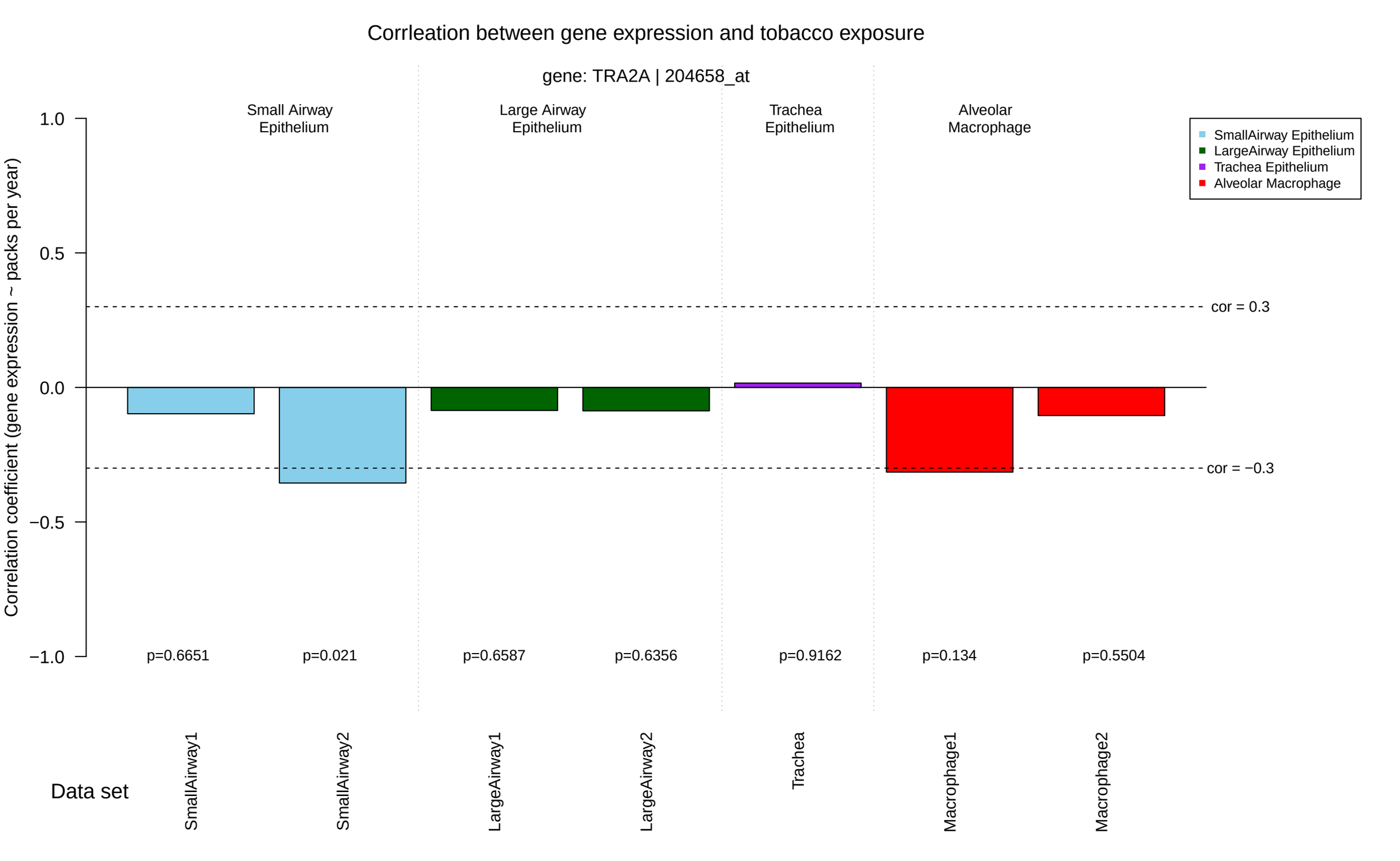

set: LargeAirway1 (p=0.6445)
TRA2A | 204658_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.049)
TRA2A | 204658_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3598)
TRA2A | 204658_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2905)
TRA2A | 204658_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8851)
TRA2A | 204658_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRA2A | 204658_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6651
p=0.021
p=0.6587
p=0.6356
p=0.9162
p=0.134
p=0.5504
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2